

PilotFish EDI SNIP Validation Processor – Complete Reference (26R1.11 WAR)

Derived from decompiled eip.war.hs.26R1.11

August 4, 2026

At a glance (plain English)

Think of this as a **HIPAA rules checker for EDI XML**. After EDI has been converted to XML, this processor runs SNIP level 1–7 validation rules (XPath-based) in parallel and returns a structured XML report of failures — either replacing the transaction body or stored as an attribute.

EDI SNIP Validation — what it does.

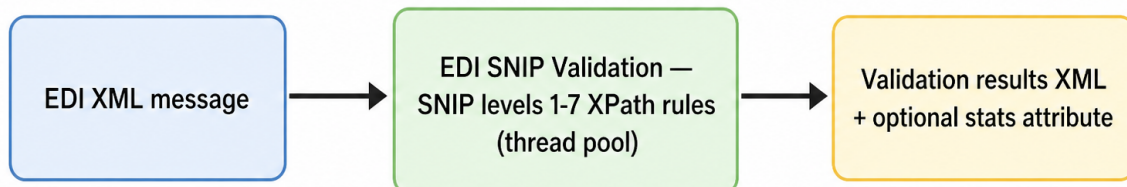


Figure 1: At a glance (plain English)

- **Validates** EDI **XML** (not raw X12 text) against selectable SNIP levels.
- **Requires** X12 version and transaction type (for example 005010X279 / 270).
- **Runs** rules concurrently via a configurable thread pool.
- **Outputs** <EdiValidationResults> XML; optional stats and attribute storage.

Purpose

This document is a **deep dive** into the **EDI SNIP Validation** processor as shipped in **eip.war.hs.26R1.11**: SNIP level selection, licensing, SNIP 5 code database setup, SNIP 7 rule files, output modes, and thread-pool validation behavior via **EdiThreadpoolValidator**.

Provenance	Value
WAR	eip.war.hs.26R1.11
Module JAR	xcs-format-edl-1.0.10.jar
Decompiler	CFR 0.152
Implementation class	com.pilotfish.eip.modules.transform.edi.EdiTransformationProcessor
Validator engine	com.pilotfish.eip.modules.transform.edi.validation.EdiValidationRules
Decompiled path	war-pipeline/decompiled/eip-26R1.11/.../EdiThreadpoolValidator.class
Processor type string	EDI SNIP Validation
Description	Performs Advanced SNIP validation on EDI XML.
Help ID	console.processor.edi_snip_validation

Related EDI stack:

Class	Role
EdiTransformationProcessor	Produce EDI XML consumed here
EdiValidationRules	Loads XPath rule sets at startup
EdiValidationCode	SNIP 5 in-memory codes database
ProductsValidator.isSnipEnabled()	License gate for SNIP 4-7 at runtime

Dialect guide: PilotFish-EDI-XML-Guide-26R1.11.md — EDI XML tree shape SNIP XPath expects (XCSDData / Loop_* / segment tags).

Derived from WAR bytecode (CFR). Descriptor uses `setImmediateSave(true)`.

1. Conceptual model

```
EDI XML stream
|
v
Optional: buffer bytes if NOT "Make errors transaction data"
|
v
License check (SNIP 4/5/7 at runtime on server)
|
v
Build EnumSet<SnipLevel> from enabled checkboxes
+ SNIP5 -> loadCodesDatabase (once per JVM)
```

```

+ SNIP7 -> loadSNIP7RuleFile path attribute
|
v
EdiThreadPoolValidator.runThreadPoolValidations(...)
|  parse XML (Saxon), load rules for EdiVersionType
|  parallel XPath evaluation per rule
v
Wrap results in <EdiValidationResults>...</EdiValidationResults>
|
+-- UseTransactionData false (default):
|   restore original stream
|   attribute com.pilotfish.edi.validation.results = results XML
|
+-- UseTransactionData true:
    replace stream with results XML (UTF-8 bytes)

```

2. High-level behavior (processData)

1. Read all SNIP boolean flags, version, document type, long names, stats, stop-on-error, use-transaction-data.
2. **Stream buffering** when UseTransactionData is false:
 - Copy stream to byte[]; reset stream to ByteArrayInputStream for validator input.
3. **License gate:** If SNIP 4, 5, or 7 enabled and ServerCore.isRunning() and not emulator and !ProductsValidator.isSnipEnabled() -> throw EIPEException("Not licensed for SNIP 4 to 7 validations during runtime").
4. Build EnumSet<SnipLevel> for enabled levels 1–5 and 7 (no SNIP 6 enum — level 7 covers 6–7 rules).
5. **SNIP 5:** loadCodesDatabase — one-time static propertyMap from Advanced tab + default implementation class.
6. **SNIP 7:** loadSNIP7RuleFile — sets attribute Snip7RuleFile to absolute path when file exists.
7. EdiThreadPoolValidator.runThreadPoolValidations(data, EdiVersionType(version, type), snipSet, bLongNames, bAddStats, bStopOnError)
8. Build results XML string in StringBuilder.
9. **Output mode:**
 - UseTransactionData false: restore original bytes to stream; set attribute com.pilotfish.edi.validation.results to results XML string.
 - UseTransactionData true: replace stream with UTF-8 encoded results XML (original payload replaced).
10. Return TransactionData.

Startup: systemStartup loads EdiValidationRules.getInstance(); failures logged but do not stop server start.

3. Configuration reference

3.1 Version and document type

UI label	Tag	Type	Required	Tooltip
X12 version to validate	SnipVersion	Multiple choice	yes	e.g. 005010X231A1
EDI transaction type to validate	EdiDocumentType	Multiple choice	yes	e.g. 270

Dynamic coupling: Changing version updates document type choices from an internal map (`docVersionHash`). Same version may map to multiple transaction types (for example 005010X279 -> 270 and 271).

Built-in version/type pairs (partial list):

SnipVersion	EdiDocumentType
005010X212	276, 277
005010X214	277
005010X217	278, 278A3
005010X216	278B1, 278B2
005010X218	820
005010X220 / X220A1	834
005010X221 / X221A1	835
005010X222 / A1 / A2	837
005010X223 / A1 / A2 / A3	837
005010X224 / A1 / A2 / A3	837
005010X279 / X279A1	270, 271
005010X231 / X231A1	999

Empty version entry exists as placeholder in the static map.

3.2 SNIP level toggles

UI label	Tag	Default	Tooltip summary
Type 1 - Integrity Testing	Snip1Validation	true	Syntax / message integrity
Type 2 - Requirement Testing	Snip2Validation	false	HIPAA implementation requirements
Type 3 - Balancing	Snip3Validation	false	Field total balancing
Type 4 - Inter-Segment Situation Testing	Snip4Validation	false	Inter-segment rules (licensed)
Type 5 - External Code Set Testing	Snip5Validation	false	External code sets (licensed)

UI label	Tag	Default	Tooltip summary
Type 6-7 - Service and Customer Specific Testing	Snip7Validation	false	Custom rules (licensed)

All level tooltips note **Requires enhanced functionality** (table data / licensed SNIP stack).

3.3 SNIP 5 Advanced tab

UI label	Tag	Type	Default
Codes database class	Snip5Default	String (required)	<code>com.pilotfish.eip.modules.transform</code>
key - values	PropertyPairs	Table	empty

SNIP 5 dependency: Property table visible only when SNIP 5 enabled.

loadCodesDatabase: Static synchronized `propertyMap` populated once per JVM from property pairs + implementation class key `Snip5_Implementation`.

3.4 SNIP 7 rule file

UI label	Tag	Type	Default
Service and Customer Rule File	Snip7RuleFile	File	empty

Visible when SNIP 7 enabled. Missing file when configured -> `EIPEXception("Unable to determine Level 6 - 7 rule file directory")`.

Runtime attribute: **Snip7RuleFile** = absolute path string on transaction.

3.5 Output and behavior

UI label	Tag	Type	Default	Notes
Include SNIP Statistics	SnipStats	Boolean	false	Adds <code>EdiValidationStatistics</code> block
Make errors transaction data	UseTransactionData	Boolean	false	Results replace stream vs attribute
Validate using long (friendly) names	SnipFriendlyNames	Boolean	false	Use long-name XPath expressions
Stop validation on Java Exception	SnipStopOnError	Boolean	true	Abort on rule engine exceptions

3.6 Inherited AbstractProcessor

Tag	Default
ExecuteProcessor	true

4. Validation engine (EdiThreadpoolValidator)

Topic	Behavior
Pool size	<code>ServerCore.getSettings().getMaxEdiPoolSize()</code> when server running; else 8
XML parse	Saxon <code>DocumentBuilder</code> on UTF-8 stream
Rule load	<code>EdiValidationRules.getInstance().getLoadedXpaths(versionType, snipLevels, data)</code>
Short names	Rules with empty short-name XPath skipped when long names false
Rule failure	Non-empty XPath match -> <code>EdiValidationRuleResult</code> with expression, params, matched values
Java exception in rule	<code>hasJavaException=true</code> ; stack trace escaped into result
Stop on error	When true, first Java exception throws <code>EIPEException</code> with escaped message
Stats block	When enabled, records pool size, timings, rule counts, success/fail totals
Success criteria	Empty result values count as successful rule

Result XML elements:

- Wrapper: `<EdiValidationResults>`
- Per rule: `<EdiValidationRuleResult exception="true|false">` with `<item>` children
- Stats: `<EdiValidationStatistics>` when requested
- XPath snippets may use `<![CDATA[...]]>` in string output

5. Transaction attributes

Key	When set	Content
<code>com.pilotfish.edi.validation.transactionData</code>	<code>UseTransactionData</code> false	Full <code><EdiValidationResults></code> XML string
<code>Snip7RuleFile</code>	SNIP 7 enabled and file resolved	Absolute path to rule file

When `UseTransactionData` true, results are on the **stream**, not the attribute above.

6. Worked example — 270 after EDI-to-XML

Route order:

1. EDI Transformation (EDI to XML)
2. EDI SNIP Validation — Version 005010X279, Type 270, SNIP 1–3 on, long names off, stats off, use transaction data **false**

Outcome: Original EDI XML restored on stream; attribute `com.pilotfish.edi.validation.results` contains failure XML for downstream branching (for example Route Condition on XPath).

7. Known quirks and caveats

1. **Input must be EDI XML** — raw X12 will not parse as expected.
 2. **SNIP 4–7 license** enforced at runtime on server (not emulator-only paths).
 3. **SNIP 5 property map is JVM-global** — first processor configuration wins until restart.
 4. **SNIP 6 not separate** — UI “Type 6-7” maps to `SnipLevel1.SNIP7`.
 5. **Default SNIP 1 on** — routes validate integrity unless disabled.
 6. **Stop on error default true** — unexpected rule exceptions halt the transaction.
 7. **Version/type must match payload** — wrong `EdiVersionType` loads incorrect or empty rule sets.
 8. **Immediate save** on configuration descriptor — UI saves SNIP processor config eagerly.
 9. **Memory logging** around SNIP 5 code DB load at debug level (MB before/after).
-

8. Operational recommendations

Goal	Guidance
Route order	EDI Transformation before SNIP Validation
Branching on errors	Keep UseTransactionData false; XPath the results attribute
Reporting	Enable Include SNIP Statistics temporarily for performance tuning
SNIP 4–7	Confirm license before enabling in production
SNIP 7 custom rules	Validate rule file path on each environment
Friendly names	Enable when maps use long element names matching schema
Fail-fast vs collect-all	Stop on error true for strict routes; false to gather all rule failures
Pool sizing	Tune server <code>MaxEdiPoolSize</code> for large rule sets

9. Source index (this WAR)

Topic	Path under war-pipeline/decompiled/eip-26R1.11/
EDI SNIP Validation processor	com/pilotfish/eip/modules/transform/edi/EdiSNIPVa
Thread pool validator	com/pilotfish/eip/modules/transform/edi/validation
Validation result XML	com/pilotfish/eip/modules/transform/edi/validation
AbstractProcessor	com/pilotfish/eip/extend/AbstractProcessor.java
Module JAR	war-pipeline/jars/eip-26R1.11/xcs-format-edi-1.0.

10. Pipeline provenance

PilotFish WARs/eip.war.hs.26R1.11

-> xcs-format-edi-1.0.10.jar

-> CFR decompile

-> Documents/Processors/26R1.11/PilotFish-EDI-SNIP-Validation-Reference-26R1.11.(md|pdf)

Deep-dive documentation from WAR decompile – EDI SNIP Validation Processor – 26R1.11 – August 2026.